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Department of Computer Science and Engineering

IV – I Project Abstract

Batch No: A04

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Domain: Artificial Intelligence and Machine Learning (AI/ML)

Title of the Project (Tentative) :

Explainable AI-Based Stock Market Trend Prediction and Risk Analysis System

Research Gap Identified :

1. **Limited interpretability of stock market prediction:**

Most existing stock market prediction methods focus on predicting stock prices or market trends using machine learning and deep learning techniques, with limited explanation of how the predictions are generated. This creates a need for Explainable AI approaches that can identify and present the important factors influencing the prediction.

2. **Limited integration of prediction and risk analysis:**

Existing approaches commonly focus on stock market trend or price prediction, while risk-related factors are often analyzed separately. This creates a need for an integrated system that can perform market trend prediction along with risk analysis to provide more meaningful information about the predicted market movement.

3. **Need for systematic explainable evaluation:**

Further investigation is required to understand the relationship between prediction performance, feature importance, and associated market risk. Comparative evaluation of prediction results and Explainable AI techniques can help in developing a more transparent and understandable stock market analysis framework.

Problem Statement:

Stock market prediction is a challenging task due to the dynamic nature of financial markets, variations in market conditions, and the influence of multiple financial factors. Existing stock market prediction approaches mainly concentrate on analyzing historical market data and generating predicted trends, while the reasoning behind the predictions may not be clearly understandable to users. The lack of interpretability can make it difficult to identify which input features have contributed to a particular prediction. Furthermore, stock market prediction and risk analysis are often considered as separate tasks, limiting the ability of users to obtain a unified understanding of predicted market trends and associated risks. Therefore, this project aims to develop an Explainable AI-based framework that performs stock market trend prediction and risk analysis while providing understandable explanations of the model's predictions. The framework analyzes relevant historical stock market information, predicts market trends using machine learning techniques, identifies important features contributing to the prediction using Explainable AI techniques such as SHAP, and provides risk-related insights for the analyzed stock.

Outcome of our project :

The proposed Explainable AI-based framework performs stock market trend prediction and risk analysis using a machine learning-based approach. The model analyzes historical stock market data and relevant financial features to identify patterns and generate predicted market trends. Explainable AI techniques such as SHAP are incorporated to identify the contribution and importance of individual features influencing the prediction. The system also provides risk-related insights based on the analyzed market information, allowing users to understand the factors associated with the predicted trend. The project provides a user-friendly interface for presenting stock market predictions, important influencing features, and risk analysis results in an understandable manner.

Abstract :

Stock market analysis involves processing large amounts of historical and financial information to identify market patterns and predict future trends. The dynamic and uncertain nature of financial markets makes accurate and understandable stock market prediction a challenging task. Existing machine learning-based prediction approaches can generate market trend predictions, but the decision-making process of these models may not be easily understandable to users. The proposed project, "Explainable AI-Based Stock Market Trend Prediction and Risk Analysis System," a develop an intelligent framework that combines stock market trend prediction, Explainable Artificial Intelligence, and risk analysis. The system analyzes relevant historical stock market data and financial features to identify patterns and predict market trends. Explainable AI techniques such as SHAP are used to identify the importance and contribution of individual features to the model's prediction, thereby improving the transparency of the prediction process.

Along with trend prediction, the system performs **risk analysis** to provide additional information about factors associated with the predicted market movement. The prediction results, feature importance, and risk-related information are presented through a user-friendly interface. The proposed system aims to provide a unified and interpretable approach to stock market analysis by combining prediction, explanation, and risk analysis with in a single frame work. By integrating machine learning with Explainable AI techniques, the system helps users understand not only the predicted market trend but also the major factors influencing the prediction and the associated risk information. This approach aims to make AI-based stock market analysis more transparent, understandable, and useful for analyzing market trends.

Keywords: Explainable AI, Stock Market Prediction, Machine Learning, SHAP, Risk Analysis, Stock Market Trend, Financial Data Analysis, Feature Importance, Artificial Intelligence, XAI.